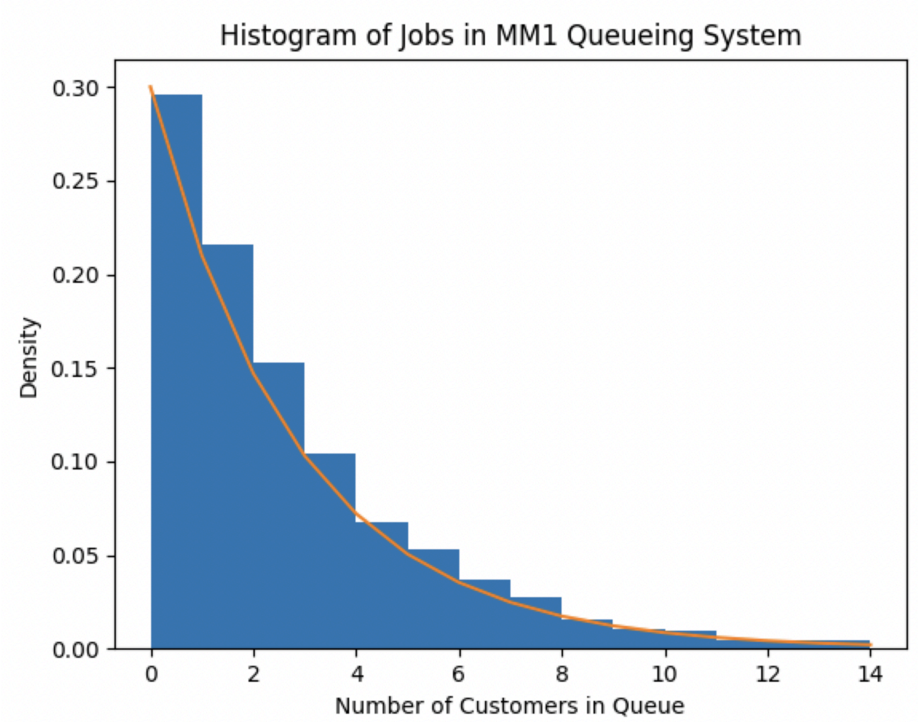
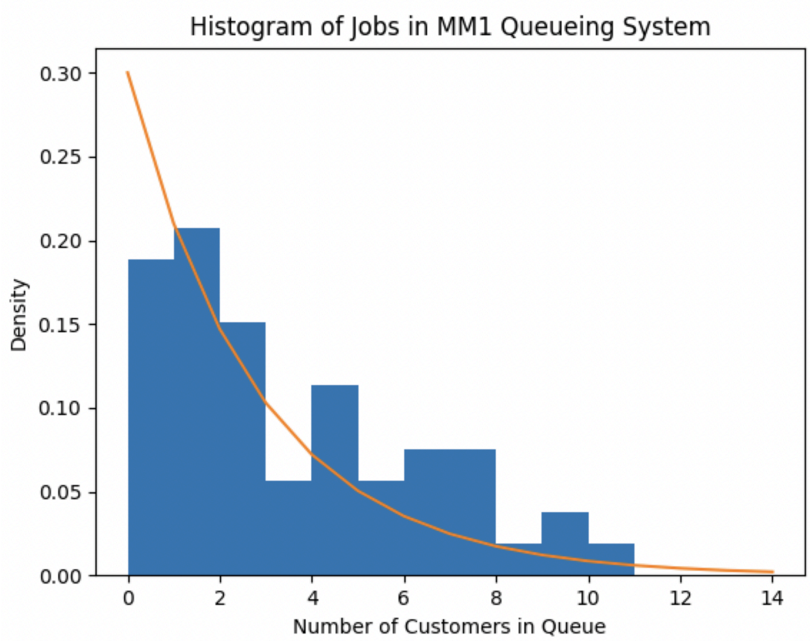
## 3.7

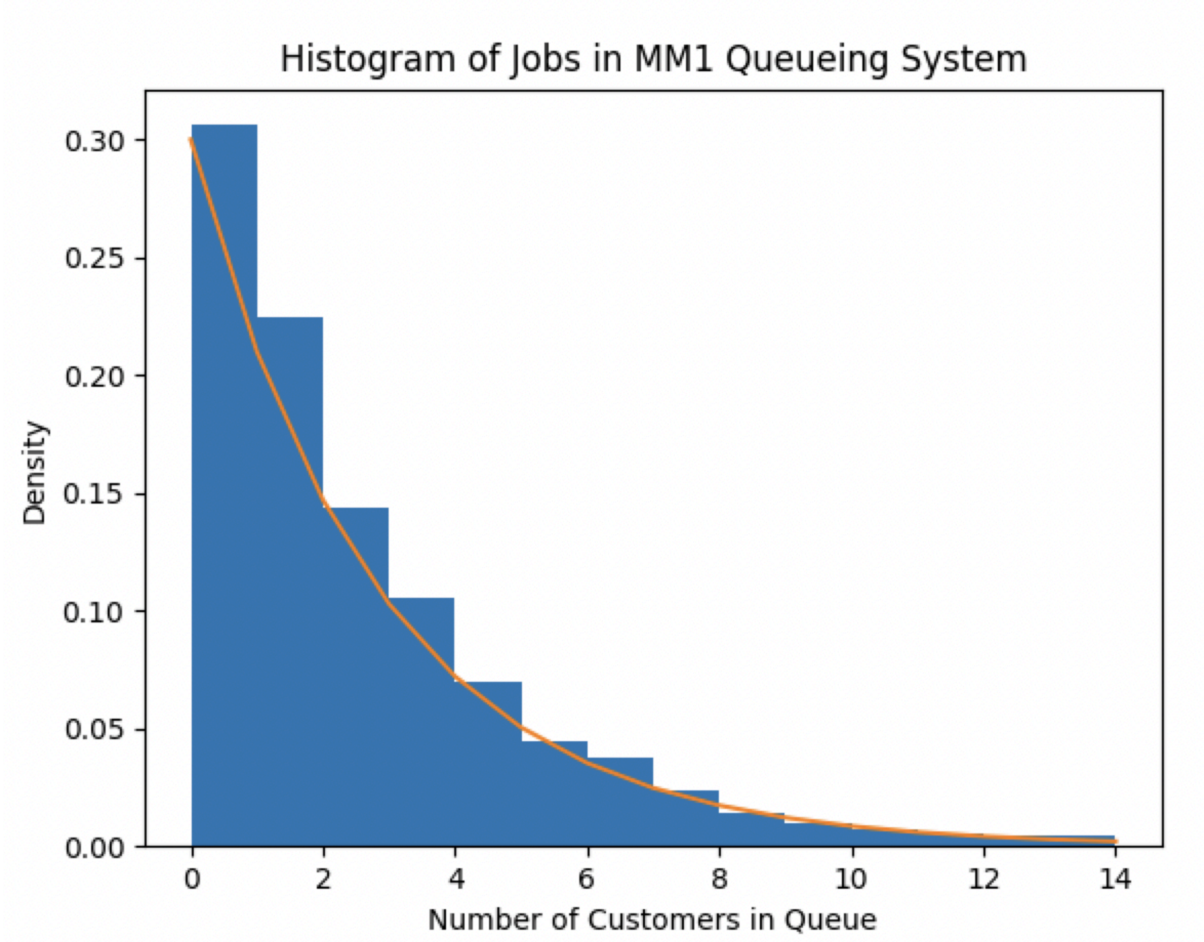
lambda = 7, mu = 10, max k = 15
The histogram for 10 000 simulated second:




The histogram for 50 simulated seconds:



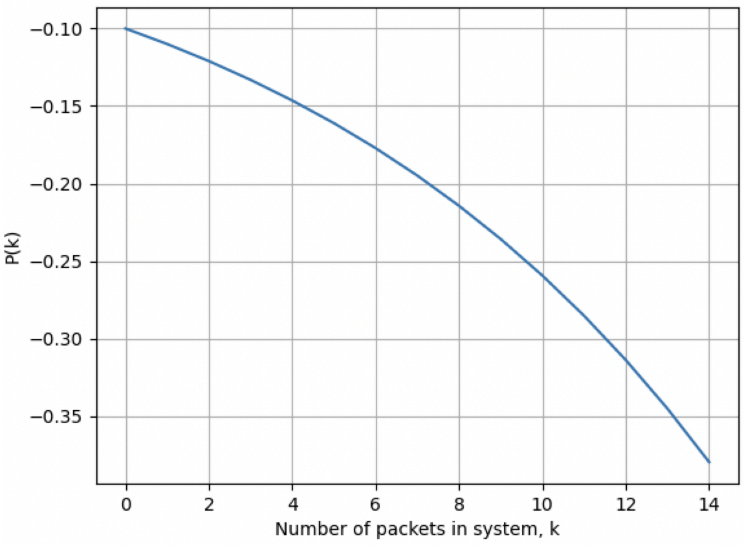
The histogram for 3000 simulated seconds:



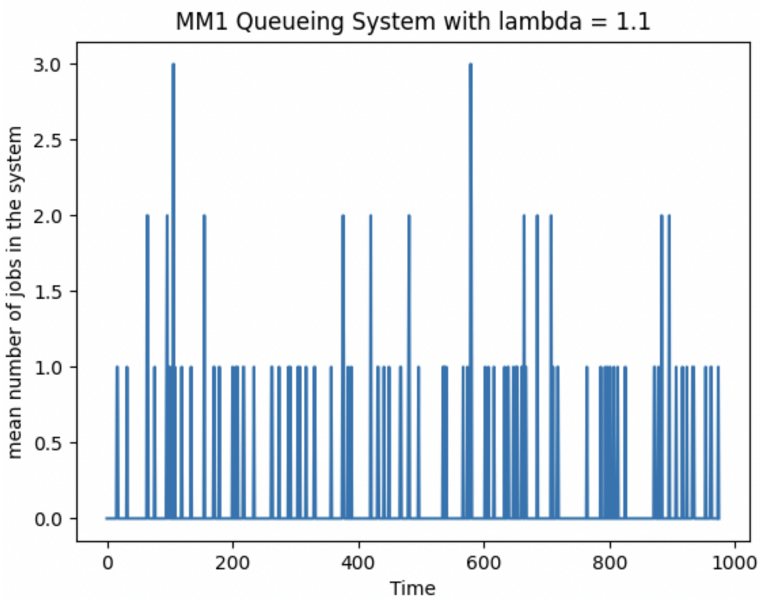

For a simulated amount of time that is too low, the accuracy decreases, because the number of measurements are fewer thus leaving room for increased fluctuations. For a longer simulation time, the mean evaluations of the measurements are more accurate because the random fluctuations (and possible outliers) average out.

## 3.8

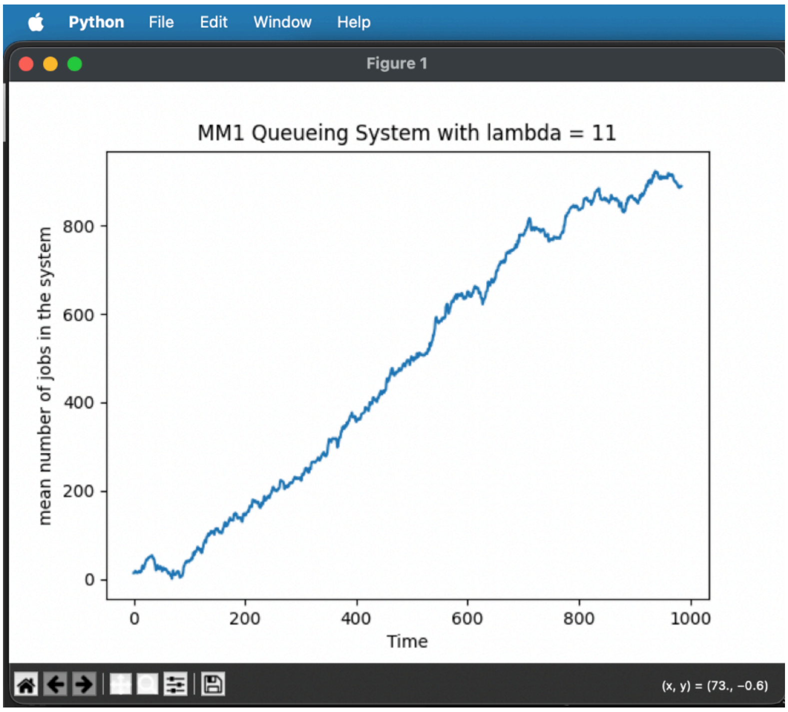
lambda = 11, mu = 10, max k = 15
for rho = 1.1



P(k)
−0.10
−0.15
−0.20
−0.25
−0.30
−0.35
0
2
4
6
8
10
12
14
Number of packets in system, k

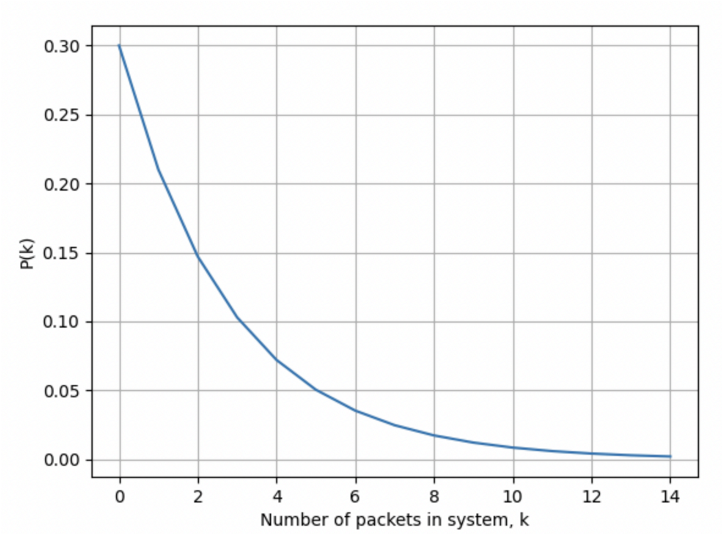



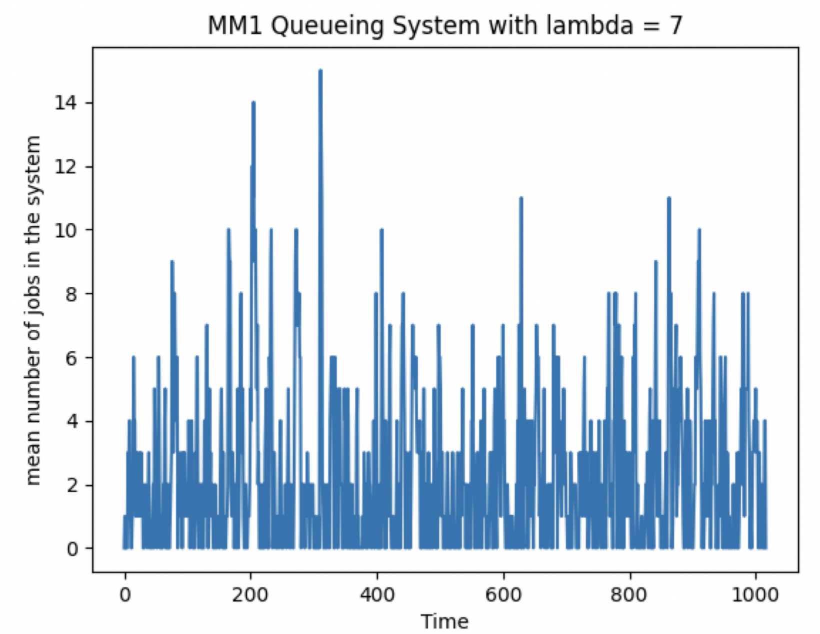
MM1 Queueing System with lambda = 1.1
mean number of jobs in the system
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
200
400
600
800
1000
Time




(might actually hit zero at some point in the beginning, however this is only possible in the beginning)

lambda = 7, mu = 10, max k = 15 (same as previous)
for rho = 0.7




When rho is 1.1, this means that the lambda is larger than mu, thus the number of calls is always increasing (increases faster than they can be served) and our queue is unstable. Thus the plot doesn't stabilize, unlike in the case for rho = 0.7, where the queue eventually reaches a stable state (i.e the server is able to process customers faster than they enter the queue)

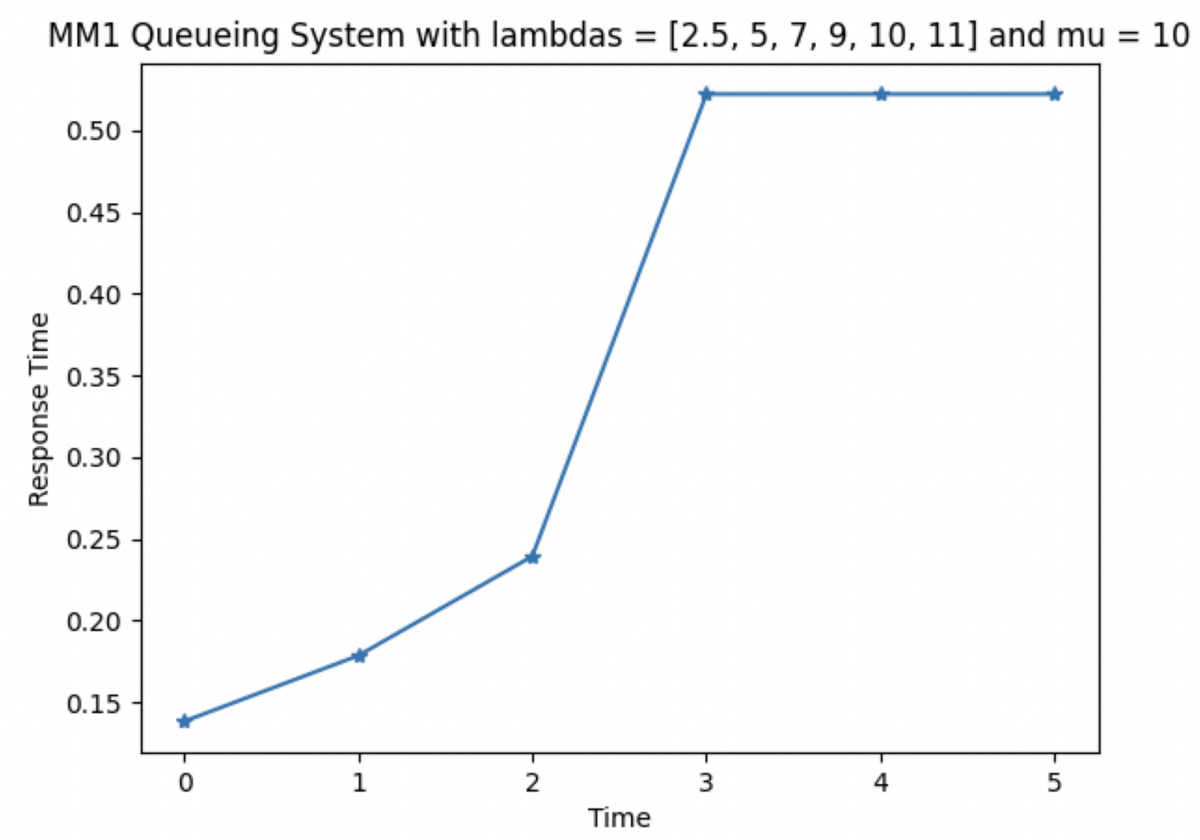
Another thing to mention is that when rho = 1.1, it is greater than 1 and per the preparation exercise (2.4) this violates the mathematical assumptions that we have made, and thus the plot makes no sense

# 3.10

lambda: 2.5, mu: 10, T: 0.1383926872737475
lambda: 5, mu: 10, T: 0.1785938084807715

lambda: 7, mu: 10, T: 0.2394177727181902
lambda: 9, mu: 10, T: 0.5221863763826906
lambda: 10, mu: 10, T: 0.5221863763826906
lambda: 11, mu: 10, T: 0.5221863763826906

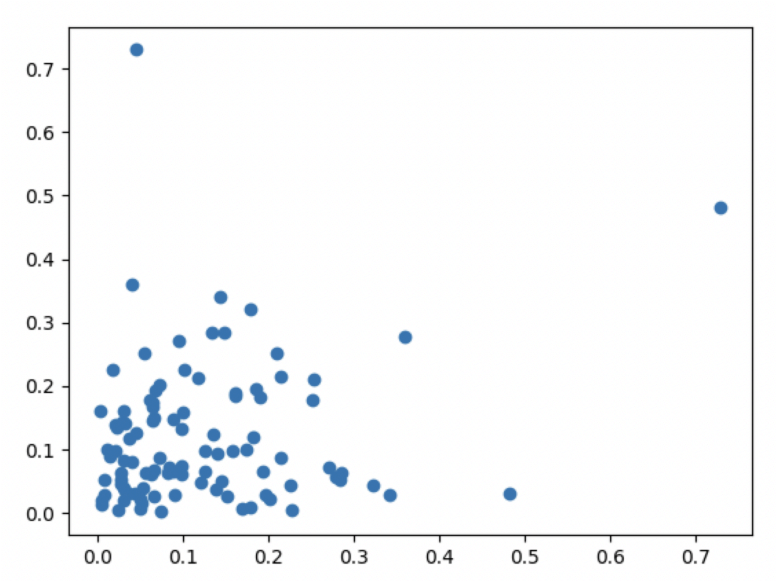
# 3.11



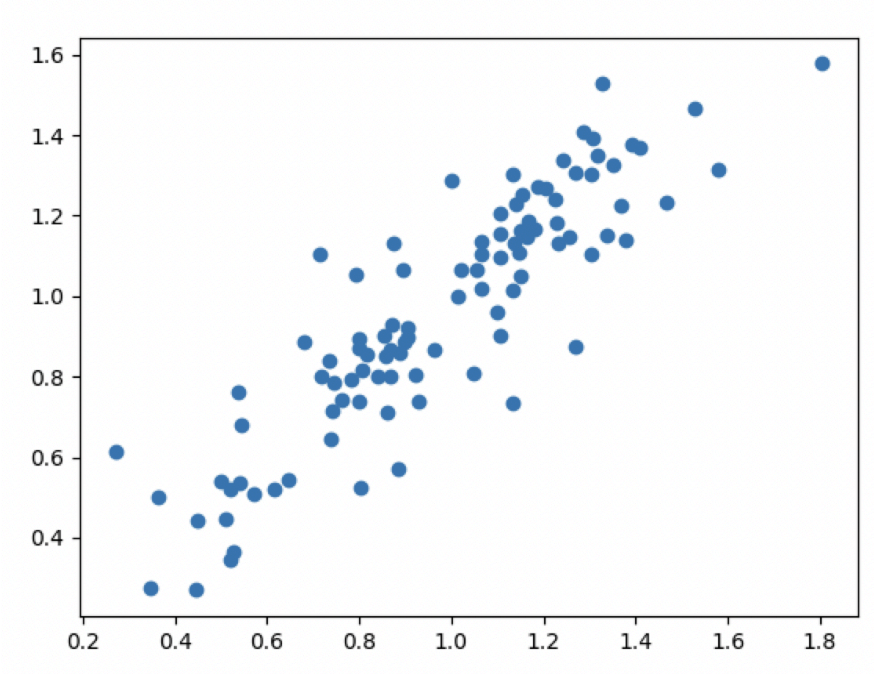

The response time increases when the system load increases

# 3.12

lambda = 2.5, mu = 10




lambda = 9, mu = 10




# 3.13

The dependency between consecutive response times increases when the load increases, this is because the number of customers entering the system increases, therefore creting congestion in the queue and increasing the response time. However when the load is low, the consecutive response time don't affect the other customers as much, because the system has sufficient time to process the incoming calls/customers

# 3.14

The initial state of an M/M/1-system is when the state is at 0, i.e number of calls/customers are 0, thus the steady state probability, p_0 is the most common.

# 3.15

For rho<1 this means that the system will eventually settle as time progresses (as shown in our plots below)
However, when rho >= 1, this means that the number of calls/customers entering the system will be greater than the number of calls the system can process per time unit, meaning that the system will never settle.

# 4.1

lambda = 1, mu = 10, simulated time = 1000
According to the kendall system the middle is the service time distribution

M/D/1
Mean number in queue:  1.7312252964426877
Mean time in queue:  0.2308763053558474

M/U/1
Mean number in queue:  1.6627906976744187
Mean time in queue:  0.2370491275666237

M/M/1
Mean number in queue:  2.564878048780488
Mean time in queue:  0.35278481246533355

M/H^2/1
Mean number in queue:  827.9719157472417
Mean time in queue:  121.21398426734436

# 4.2

Increasing the variance of the service time increases both the mean number of jobs in the system E[N] and the mean response time T

## 4.3

The system becomes unstable when rho ≥ 1, thus when lambda ≥ mu is the system unstable

## 5.1

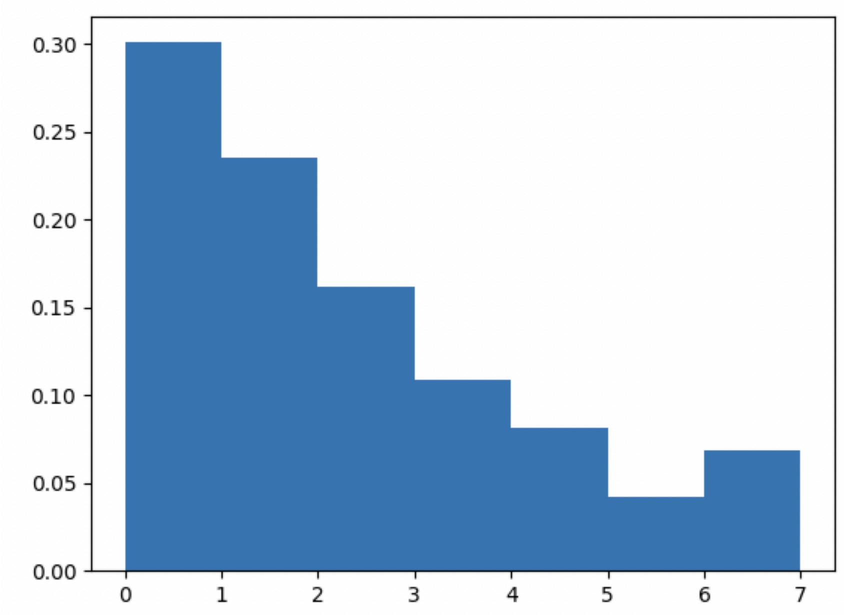
7 jobs can now coexist in the system, i.e 1 being served and 6 in the buffer.

## 5.2

I got that the blocking probability is:
Blocking probability:  0.026681805279591256

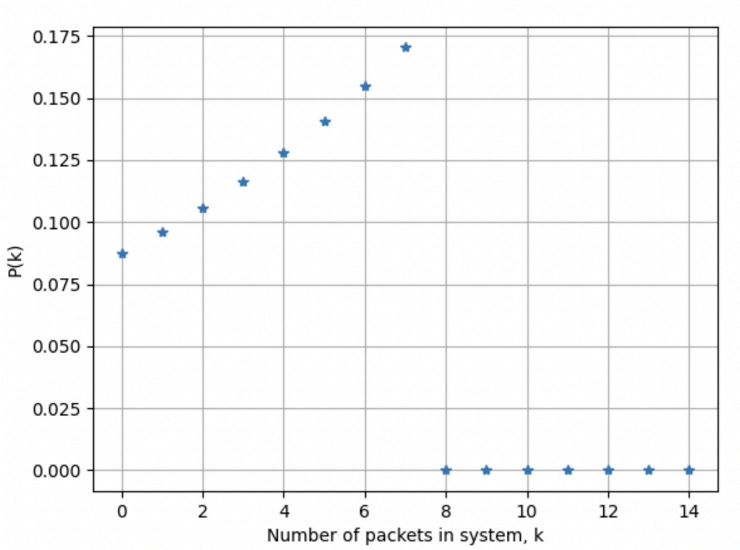
## 5.3

Probability system is full: 0.026262626262626262




## 5.4

in pkMM1b.py
Arrival rate (lambda): 11
Service rate (mu): 10
Buffer size: 6
Maximum k value to plot: 15




System probability of being full: 0.1704036523407395 = congestion rate

## 5.5

In infinite queues, high load causes instability and unbounded queue growth. In finite-buffer systems, the queue length remains bounded because excess arrivals are blocked, so increased load results in higher congestion rate rather than instability.